

Seeyan Newaz

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Education

The University of Texas at Dallas

Aug 2022 – May 2026

Bachelor of Science in Computer Science

GPA: 3.92

- **Relevant Coursework:** Computer Architecture, Discrete Maths, Data Structures & Algorithms, Software Engineering, Programming Languages & Paradigms, Probability & Statistics, Database Systems, Linear Algebra, Computer Networks, Digital Logic, Automata Theory
- **Organizations:** Association for Computing Machinery at UTD (Member), Artificial Intelligence Society at UTD (Member), Bangladeshi Student Organization at UTD (Member)
- **Achievements:** Academic Excellence Scholarship, Texas Competitive Scholarship, Dean's List, Edexcel High Achievers' Award, The Daily Star Award

Technical Skills

Languages: Python, Java, C/C++, C#, PHP, HTML/CSS, JavaScript/TypeScript, Bash, SQL, Assembly, Dart, Lisp, Prolog, VHDL

Frameworks/Libraries: React.js, Next.js, Node.js, Angular, Spring Boot, .NET, Flask, Tailwind CSS, SQLAlchemy, Flutter, Pandas, NumPy, scikit-learn, Matplotlib, D3.js, Chart.js, SciPy, MNE

Developer Tools: Git, Docker, Kubernetes, Linux, Postman, JIRA, Hugging Face, OpenAI API

Cloud & Databases: AWS, Microsoft Azure, PostgreSQL, MySQL, Firebase, Oracle

Experience

Machine Learning Engineer

Jan 2026 – May 2026

The University of Texas at Dallas

Richardson, TX

- Built a reproducible **Python EEG classification pipeline** using **MNE**, **SciPy**, **pandas**, & **scikit-learn** for Alzheimer's detection.
- Implemented **Welch PSD** feature extraction across **19** EEG channels, deriving Delta, Theta, Alpha, Beta, & ratio-based biomarkers for downstream classifiers.
- Engineered posterior-channel, log-transformed, & derivatives-based features, improving **Random Forest accuracy to 80.0%** & **ROC-AUC to 0.861**.
- Validated **Logistic Regression & Random Forest** with subject-level **5-fold CV**, preventing data leakage & improving reliability of participant-level evaluation.

Peer Tutor for Calculus & Linear Algebra

Aug 2023 – May 2026

Student Success Center – The University of Texas at Dallas

Richardson, TX

- Achieved **60%** average grade improvement over **500+** undergraduates by delivering personalized 1:1 tutoring.
- Enhanced concept mastery & confidence of students by tailoring explanations & practice to diverse learning styles.
- Boosted exam readiness & achieved **100%** student satisfaction by creating custom problem sets & step-by-step solutions.

Software Developer Intern

May 2025 – Aug 2025

Paycom

Irving, TX

- Engineered a full-stack **C#/.NET** PDF data extraction pipeline for new client onboarding, projected to generate **\$100K+** savings.
- Cut manual setup by up to **5** days per client by building a **React**-based annotation tool & intelligent table-detection algorithms.
- Reduced manual labeling/review time by **70%** by developing custom **NLP/NER** models for structured entity extraction.
- Designed inference & auto-classification backend services, enabling automated data mapping & speeding up downstream analytics.

Backend Software Developer

Feb 2024 – Apr 2024

Association for Computing Machinery – The University of Texas at Dallas

Richardson, TX

- Built a **Dart/Flutter** events app for **30K+** students across **50+** university organizations by working in a **Scrum-based 5-person** team.
- Delivered **90%** accurate personalized event recommendations by integrating a **GPT-4 Turbo** chatbot into the user experience.
- Enhanced data retrieval efficiency by **50%** by designing a **Firestore** schema for real-time synchronization.

Projects

Trace | Python, Pandas, NumPy, scikit-learn, Matplotlib, yfinance, D3.js, Chart.js

Apr 2026

- Built a **post-earnings stock prediction platform** at FinHack 2026 for **64** earnings events across **8** AI-linked companies using market & text signals.
- Created company, peer, regulation, & event-level sentiment features from **news, transcripts, press releases, & video data**.
- Engineered a leakage-safe **NLP pipeline** with **7-day** windows, spillover features, AI keyword counts, & conflict indicators.
- Raised test **accuracy** from **50.0%** to **62.5%** & **ROC-AUC** from **0.667** to **0.792** using chronological model evaluation.
- Built an interactive **D3.js/Chart.js** dashboard with model comparison, contagion mapping, case studies, & feature-importance views.

LedgerLens | Docker, Spring Boot, Angular, PostgreSQL, OpenAI API

Feb 2026

- Developed an **AI-driven, full-stack** web app using **GPT-5.2** that enables users to upload receipt images & track expenditures.
- Cut manual receipt processing by **50%** through automated extraction & analysis of expenditure data.
- Built a dynamic **Angular** frontend that displays all-time/monthly insights, real-time data visualization & spending trends.
- Implemented a scalable backend with **Spring Boot**, **PostgreSQL**, & **Docker** for persistent storage & consistent deployment.

InboxIntel | Python, Streamlit, PostgreSQL, Gmail API, OpenAI API

Nov 2025

- Engineered an **AI-powered, full-stack** intelligent inbox assistant using **GPT-5.1** to retrieve & process unread messages in real time.
- Cut manual email processing time by **60%** through automated summaries, prioritization, task extraction, & reply drafts.
- Designed a responsive **Streamlit** interface by building dynamic filters & metrics.
- Reduced duplicate processing by implementing **PostgreSQL** state management & deduplicated query flows.

HydraWatch | Flask, Next.js/React.js, SQLite, SQLAlchemy, ML

Nov 2024

- Built a **full-stack** web app for EOG Resources at HackUTD 2024 to detect hydrate anomalies in gas pipelines in real time.
- Achieved **95%** model accuracy by training a regression model in Google Colab & deploying it via **Flask** for live predictions.

InternTrack | Flask, React.js, SQLite, SQLAlchemy

Oct 2024

- Developed a **full-stack** web app with **CRUD** functionalities & a **SQLite** backend to manage & track internship applications.
- Implemented real-time status updates & a responsive **Chakra UI** design to ensure accurate record-keeping across all devices.